

Hotel Management System

Project Report

1. Introduction

The Hotel Management System is a basic Python program for handling rooms and bookings at VIT's Paradise hotel. It runs in a terminal and uses a menu for its different tasks.

2. Aim

The aim of this project is to show room availability, record bookings, calculate bills, find guest details, and manage check-outs.

3. Tools and Python Concepts

The program uses Python, dictionaries to store room and booking information, functions for each task, and a loop and conditions to control the menu.

4. Room Information

The program lists six rooms: 101, 102, 103, 201, 202, and 301. Each room has a type, occupancy limit, price, food charge, extra bed charge, status, facilities, and complimentary items.

5. Main Menu Features

- 1. Available Rooms:** Shows rooms marked as available.
- 2. Room Details:** Shows information about all rooms, including prices, occupancy, facilities, and complimentary items.
- 3. Book Room:** Collects booking information, lets the user choose food and an extra bed, calculates the bill, and marks the room as booked. A booking can be made for up to three people.
- 4. Guest Details:** Shows saved booking details when a room number is entered.
- 5. Booked Rooms:** Shows the first booked room it finds, including the guest name.
- 6. Generate Bill:** Shows room, food, and extra bed charges and the total.
- 7. Check-out:** Shows the final bill. If checkout is confirmed, it makes the room available and removes the booking.
- 8. Exit:** Displays a thank-you message.

6. Bill Calculation

The room price is multiplied by the number of nights. If food is included, its daily charge is multiplied by the number of nights. An extra bed charge is added when an extra bed is selected and available. The program adds these charges to calculate the total.

7. How to Run

Save the code as a Python file. Open a terminal in the folder containing the file and run it with Python. The menu will then appear.

8. Limitations

- Booking information is stored only while the program is running. It is not saved to a file or database.
- The program asks for details for each person but stores only one person's details for a room.
- The booked rooms option shows only the first booked room it finds.
- The Exit option displays a message, but the menu continues to run.

9. Conclusion

This project demonstrates basic Python functions, dictionaries, loops, conditions, and user input. It provides a simple way to view rooms, make bookings, calculate bills, and check guests out.